

Pink Edge AI — Technical Refinement & Mammography Model Report

1. Purpose

This report combines the current technical audit with a focused review of possible pretrained mammography models.

The goal is **not to redesign Pink Edge AI**, but to:

- identify the most serious current weaknesses
 - determine what should be fixed before the hackathon
 - verify what is actually working
 - identify a realistic mammography model if the current one cannot be integrated
 - avoid unsupported technical or medical claims
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2. Current Main Problem

The prototype UI and workflow are implemented, but the current `Run Triage` path does **not yet demonstrate genuine mammography model inference**.

The application can display:

- malignant/benign result
- confidence score
- latency
- BI-RADS
- ACR density

but the current code generates predefined/randomized results instead of clearly loading a trained mammography model and producing the result from the uploaded image.

Therefore, the current priority is:

Image → Real trained model → Real inference → Real output

rather than adding more UI features.

3. Highest-Priority Problems

Problem	Why it matters	Priority
Genuine mammography inference not integrated	Core AI claim cannot currently be verified	5/5
Random confidence/results	Cannot be presented as genuine model output	5/5
Model .pt files need verification	File existence does not prove trained usable weights	5/5
Accuracy claims unsupported	Published/model confidence $\neq$ measured accuracy	5/5
BI-RADS/ACR output source unclear	Could be interpreted as direct clinical AI diagnosis	5/5
RK3588/NPU execution not proven	UI telemetry is not hardware evidence	4/5
GSM transmission not proven	“Broadcasted” on screen does not prove physical transmission	4/5
Alibaba Cloud integration not proven	Current implementation includes simulated cloud operations	4/5
Maternal model not verified	<code>fetal_health.pt</code> needs actual model/inference verification	4/5

4. What Must Be Fixed First

A. Genuine AI inference

The software team should make at least **one real model inference path** work.

Required chain:

Mammography image $\rightarrow$ preprocessing $\rightarrow$ trained model $\rightarrow$ inference $\rightarrow$ prediction $\rightarrow$ UI

The displayed confidence and detection must come from that inference.

Priority: 5/5

B. Verify existing models

The repository currently contains:

- yolo8_obb_int8.pt
- tb_classifier.pt
- fetal_health.pt

These should be checked to determine:

- whether they contain genuine trained weights
- what dataset they were trained on
- what task they perform
- how they are loaded
- whether they actually run inference

The TB model should receive particular attention because the team has said it is already completed.

Priority: 5/5

C. Remove misleading simulation

Randomized medical outputs and dummy model generation should not remain as if they are genuine AI.

If a component is still simulated, it should be clearly identified as:

Simulated Prototype

rather than presented as completed functionality.

Priority: 5/5

5. Mammography Model Research

The current Pink Edge design is based around YOLO-style detection, so a replacement model should not be selected simply because it has a high reported accuracy.

It needs to be:

- genuinely pretrained
- realistically testable
- supported by usable code/weights
- compatible with the existing workflow

- reasonably suitable for edge deployment
- legally usable for the intended hackathon demonstration

Candidate 1 — VinDr-Mammo YOLOv5-S

Best candidate to test first.

Dataset

VinDr-Mammo.

The dataset contains 5,000 mammography exams / 20,000 images with four standard views per exam.

Task

Mammographic lesion/mass detection. The public implementation also investigates BI-RADS-related detection.

Architecture

YOLOv5-S / YOLOv5-L.

Input

Mammography images prepared for YOLO detection.

Weights & code

The implementation provides trained YOLO variants and inference/training code.

Model size

YOLOv5-S is substantially smaller than the large YOLOv5 variants, making it a more realistic edge candidate.

License

Needs verification before reuse.

The underlying YOLOv5 project has licensing considerations, so the exact combination of code, weights and intended use must be checked.

RK3588 suitability

Good candidate for testing.

YOLO-style models are much closer to the current Pink Edge architecture and can potentially be optimized for edge inference.

Main limitation

The model should **not** automatically be described as producing the complete BI-RADS 0–6 clinical assessment currently shown in Pink Edge.

Assessment: ☐ **Best practical candidate to investigate first.**

6. Candidate 2 — PHBreast PHResNet18

A lightweight mammography classification approach using multi-view mammography information.

Dataset

CBIS-DDSM and INbreast.

Task

Breast-cancer classification.

Architecture

PHResNet18/50 and other hypercomplex architectures.

Model size

The project reports relatively small variants, including approximately:

- PHResNet18: 13M parameters / ~50 MB
- PHYSBOnet: 7M parameters / ~27 MB

Weights & code

Pretrained weights and inference/training code are available.

License

MIT.

RK3588 suitability

Potentially good because of the smaller model variants, but actual export and benchmark testing would still be required.

Main limitation

This is **not a drop-in replacement** for the current single-image YOLO workflow.

It is designed around multiple mammography views, so the current prototype workflow would need adjustment.

Assessment: ☐ **Good technical candidate, but less convenient for the current prototype.**

7. Candidate 3 — DDSM / INbreast ResNet50 or VGG16

Older research models that provide pretrained mammography classification approaches.

Dataset

DDSM / INbreast.

Task

Whole-image breast-cancer classification.

Architecture

ResNet50, VGG16 and other CNN approaches.

Reported performance

The published implementations report AUC values in the approximate **0.86–0.96 range**, depending on dataset, architecture and evaluation setup.

These numbers **must not be presented as Pink Edge's accuracy**.

Weights & code

Available for several models.

RK3588 suitability

Possible after conversion/optimization, but less naturally aligned with the current YOLO/NPU direction.

Main limitation

These models provide classification rather than the lesion-localization workflow currently represented by Pink Edge.

Assessment: ☐ Useful fallback for genuine classification, but not the preferred option.

8. Candidate 4 — Mammo-FM

A newer breast-specific foundation model.

Strength

Technically powerful and interesting for research.

Problem for Pink Edge

It is substantially heavier and more complicated than what is needed for the hackathon.

The model environment uses large PyTorch/CUDA requirements, and the model weights are much larger than the lightweight edge candidates.

There are also significant restrictions around the model weights' permitted use.

RK3588 suitability

Poor for the current hackathon goal.

Assessment

● **Do not use this as the immediate Pink Edge mammography model.**

High research capability does not make it the right engineering choice for this project.

9. Model Comparison

Model	Task	Real weights	Code	Model size	Single-image workflow	Edge potential	Pink Edge fit
VinDr YOLOv5-S	Detection	✓	✓	Small	✓	□	★★★★★
PHBreast PHResNet18	Classification	✓	✓	Small	✗ Multi-view oriented	□/□	★★★★★
DDSM/INbreast ResNet/VGG	Classification	✓	✓	Medium/Large	✓	□	★★★★★
Mammo-FM	Foundation/downstream tasks	✓	✓	Very large	Complex	●	★★★★★

10. Important Accuracy Warning

Published research performance should **not** be copied directly into Pink Edge's presentation.

For example:

Model AUC = 0.95

does **not** mean:

Pink Edge accuracy = 95%.

To make our own performance claim, we would need an appropriate evaluation dataset with ground-truth labels and calculate relevant metrics such as:

- sensitivity
- specificity
- precision
- recall
- F1
- AUC
- mAP where appropriate

Until that evaluation is performed, use wording such as:

“Pretrained research model being evaluated for integration.”

rather than claiming a specific Pink Edge accuracy.

11. Recommended Decision

Do **not** immediately replace the existing YOLOv8-OBB architecture.

First verify whether:

`yolov8_obb_int8.pt`

is a genuine trained mammography model and whether the software team can load and run it.

If the existing YOLOv8-OBB model is genuine and usable:

Keep it.

This is likely the fastest path because the UI and workflow are already designed around YOLO detection.

If the existing mammography model is not usable:

Test VinDr-Mammo YOLOv5-S first.

It is the closest practical alternative to the current architecture.

If detection is not required:

PHBreast or a DDSM/INbreast classification model could be considered, but they would require more changes to the existing workflow.

12. What Should Be Demonstrated Before the Hackathon

The strongest realistic target is not three perfect medical models.

It is:

1. One genuine model

Real image → real inference → real output.

2. Evidence

Show the actual model weights, inference code and test result.

3. Honest performance

Only report metrics that have actually been evaluated.

4. Clear separation

Mark components as:

- **Implemented & verified**
- **Implemented but simulated**
- **Planned**

5. Hardware evidence

If RK3588/NPU is actually working, demonstrate it with real execution/latency evidence.

13. Final Recommendation

The project should **not become bigger just for the sake of appearing more advanced.**

The most valuable refinement is to make the existing core claim credible:

Pink Edge AI can take a mammography image and genuinely perform local AI inference rather than displaying simulated results.

For the mammography model specifically:

1st choice: Verify and integrate the existing YOLOv8-OBB model.

Fallback: Test VinDr-Mammo YOLOv5-S.

Alternative: PHBreast or DDSM/INbreast classification models if the team decides detection is not essential.

Avoid for this hackathon: Mammo-FM.

The next technical decision should therefore be based on **actual testability and integration**, not benchmark accuracy alone.